

P001-MVB

Responsible AI Clinical Decision Support Prototype

Explainable and Responsible AI for Differentiating Bacterial and Viral Meningitis: A Clinical Decision Support System

Presenter

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Accurate Prediction

Explainable AI

Programme

3MTT DeepTech Cohort 2 –
DS/ML Mentorship

Location

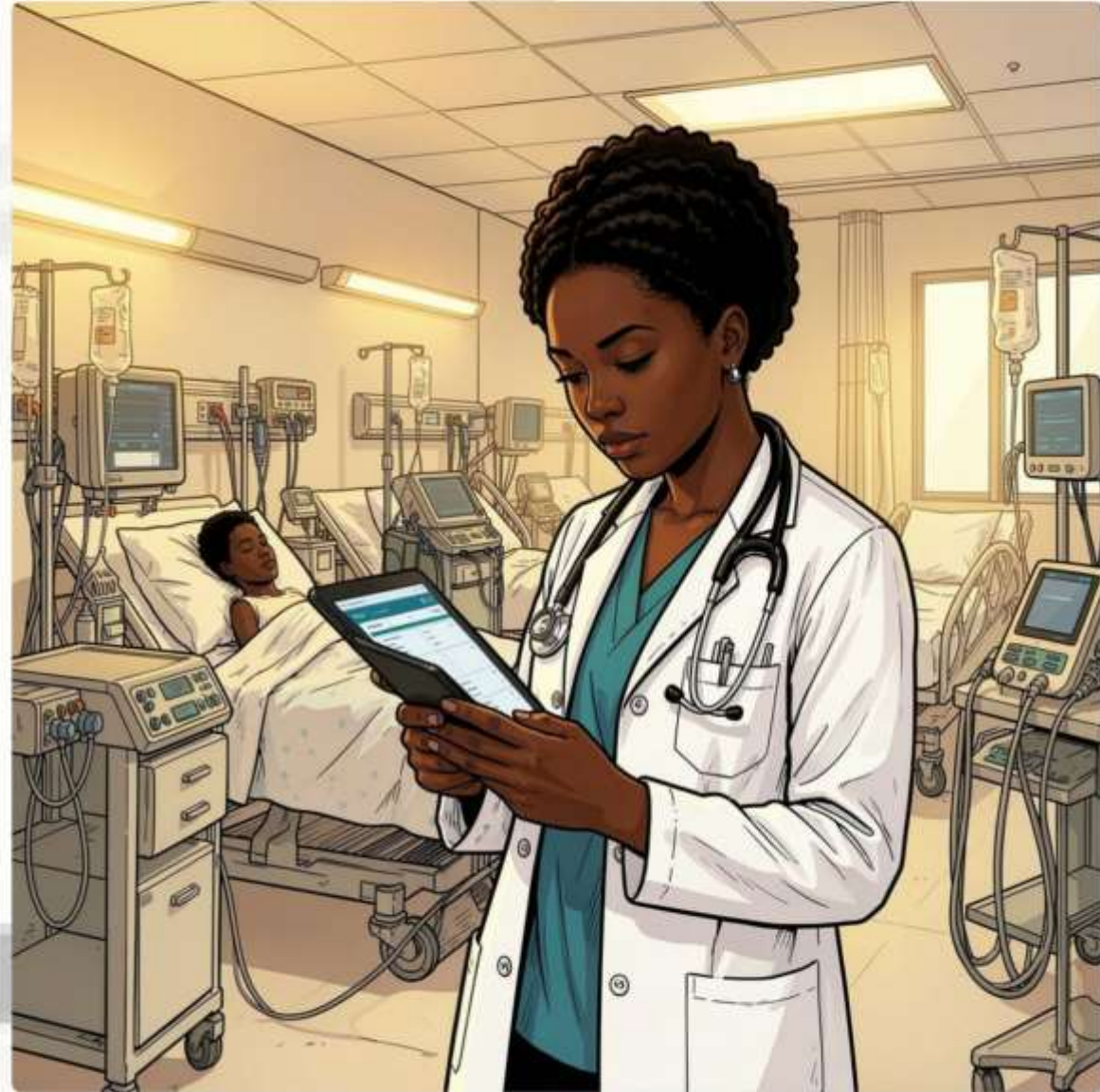
Abuja, Nigeria

Responsible
Governance

Human Oversight

The Problem

- ⚠️ **Delayed Diagnosis** — Rapidly distinguishing **bacterial** from **viral meningitis** is clinically challenging, yet treatment decisions must often be made **urgently**.
- ⚠️ **Black-Box AI** — Many healthcare AI systems operate as **black boxes**, offering little explanation for their predictions.
- ⚠️ **Limited Transparency** — This lack of transparency reduces clinician trust and confidence, increases automation bias, and limits safe adoption in healthcare.



Our Solution — P001-MVB

P001-MVB integrates four complementary components delivered through an interactive **Streamlit** clinical decision-support application, prioritising transparency, explainability, governance, and mandatory human oversight.



Explainable Machine Learning

Logistic Regression with interpretable outputs



SHAP Explanations

Feature-level transparency for every prediction



Responsible AI Governance

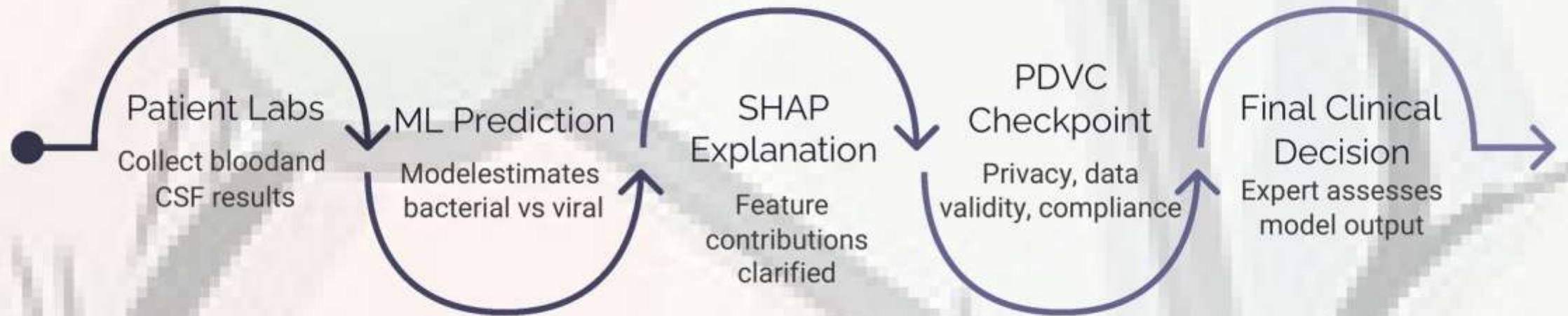
Model Card, AI Risk Assessment, structured governance



PDVC

Pre-Decision Verification Checkpoint — mandatory clinician review

P001-MVB Clinical Decision Workflow



Every prediction passes through explainability and a mandatory verification checkpoint before influencing any clinical decision – supporting, not replacing, clinical judgement.

Tools & Technologies

Programming & ML

Python · Pandas · NumPy · Scikit-learn
· XGBoost · SHAP

Deployment

Streamlit Community Cloud · GitHub

Development

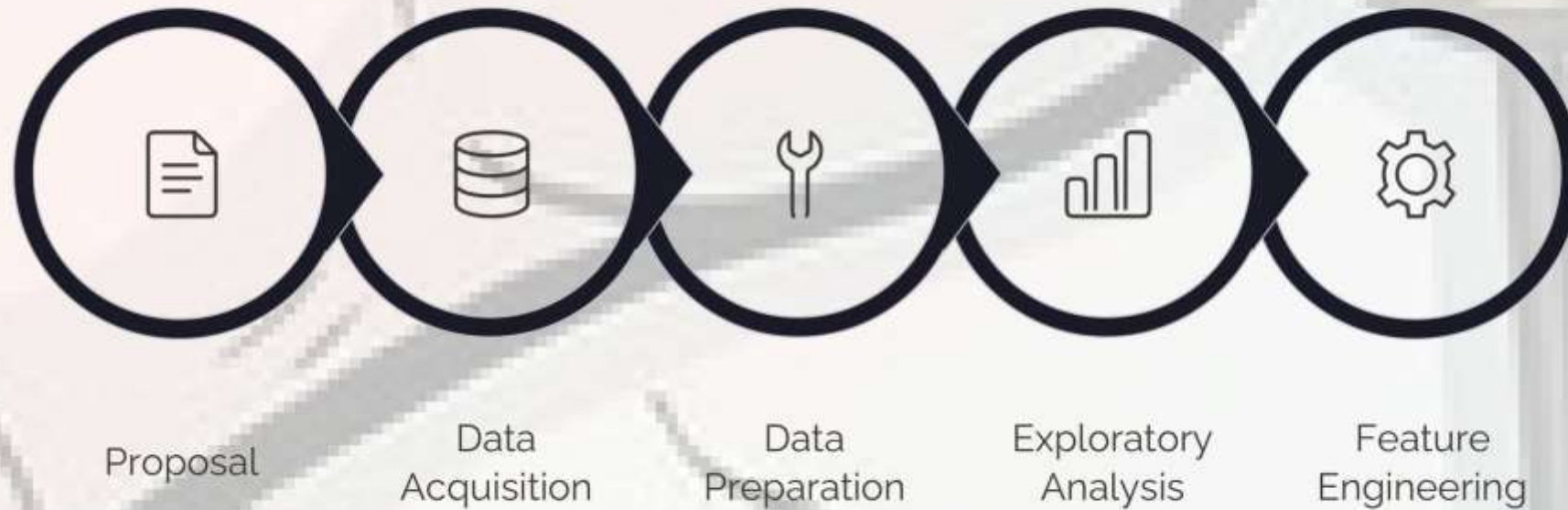
Google Colab · Jupyter Notebook

Responsible AI

Model Card · AI Risk Assessment ·
PDVC Framework



End-to-End AI Development Lifecycle



Six machine learning models were evaluated. The final model was selected based on **Bacterial Recall** – prioritising patient safety over overall accuracy.

Near-Proxy Removal

Removed **Pathogen_Present** feature to prevent data leakage

Feature Set Comparison

Full vs Restricted feature sets rigorously evaluated

Safety-First Model Selection

Bacterial Recall prioritised – minimising missed diagnoses

Responsible AI Governance

Embedded governance integrated directly into the deployed application

Model Development & Selection

Final Selected Model: Logistic Regression (Restricted Feature Set)

Dataset

Source: Kaggle – Meningitis Classification (ChanTest)

Original records: 1,200

Removed: 67 (“Unknown” diagnosis)

Final records: 1,133

Bacterial: 595 · Viral: 538

Models Evaluated

Logistic Regression — Full / Restricted

Random Forest — Full / Restricted

XGBoost — Full / Restricted

Selected: Logistic Regression (Restricted)

Best balance of bacterial recall, interpretability, transparency and Responsible AI deployment requirements.



Live Application Demonstration

Final Selected Model: Logistic Regression (Restricted Feature Set)

Streamlit Home

MVB Meningitis Classifier

Download and Runpage 12 of 16: [Download and Runpage 12 of 16](#)

⚠️ EXTENDED FUTURE - NOT FOR CLINICAL USE This tool has not undergone clinical validation, regulatory review, or certification. It must never be used for the subsequent diagnosis or treatment decisions. See Full Report

Learn More

Report Information

Patient Laboratory Values

Age (years)	CRP (mg/L)	ESR (mm/hr)
65	50	100
Gender	CRP (mg/L)	Protein
Female	50	100
CRP (mg/L)	Protein (g/L)	CRP Level
50	100	50

Click on any of the values to toggle negative or positive control

Generate Report

The meningitis classifier has been designed for the purpose of educational use only. It is not a medical device and should not be used for clinical diagnosis or treatment decisions. See Full Report

Prediction Result



SHAP Waterfall



PDVC Verification



Results & Responsible AI Outcomes

93.4%
Accuracy

Overall model accuracy on held-out test data

95.8%
Bacterial Recall

Critical safety metric — minimising missed bacterial cases

91.9%
Precision

Proportion of bacterial predictions that are correct

5

PDVC Triggers

Automated governance triggers plus clinician-initiated escalation

Responsible AI Outcomes

- SHAP provides transparent **feature-level explanations** for every prediction
- All predictions include **confidence scores**
- All five false-negative evaluation cases would have **triggered PDVC verification** before influencing clinical decisions

✅ All five false-negative cases would have been flagged by the PDVC — demonstrating the safety net in practice.

Live Application: Screenshots of the Streamlit Home Screen, Prediction Results, SHAP Explanation, and PDVC Verification screens are available in the live demonstration.

Pre-Decision Verification Checkpoint (PDVC)

Many healthcare AI projects stop after producing an accurate prediction. P001-MVB goes further by integrating Responsible AI governance directly into the clinical decision-support workflow.

Detects

Higher-risk predictions automatically

Requires

Mandatory clinician verification before action

Promotes

Transparency through embedded explainability

Reduces

Automation bias and over-reliance on AI

Reinforces

Clinician authority and accountability

```
pdvc_triggers.py (excerpt)

# PDVC trigger checks (excerpt -- full function on GitHub)
if confidence < 0.60:
    triggers.append(("Low model confidence", ...))
if clinician_override:
    triggers.append(("Clinician-requested verification", ...))
if missing_or_invalid:
    triggers.append(("Data quality validation failure", ...))
```

Governance is not documentation—it is an operational safeguard embedded in the deployed AI system.

Conclusion & Future Work

→ Key Takeaway 1

Explainable AI improves trust by making each prediction transparent to clinicians.

→ Key Takeaway 2

Safety-first model selection prioritises bacterial recall to reduce missed critical cases.

→ Key Takeaway 3

Responsible AI governance is embedded directly into the workflow through PDVC.

→ Key Takeaway 4

Human oversight remains central, supporting clinical judgement rather than replacing it.

01

Expand Dataset Coverage

Collect additional clinical cases to improve robustness and generalisability.

02

Validate Across Sites

Test performance with data from multiple hospitals and care settings.

03

Refine Governance Rules

Enhance PDVC triggers, thresholds, and escalation logic for safer deployment.

04

Improve Model Performance

Compare additional interpretable models and calibration strategies.

05

Prepare Clinical Integration

Develop workflow-ready deployment features for real-world use.

Limitations

Research prototype only — not intended for clinical use. · Dataset is clinically inspired rather than clinically validated.

No external validation has yet been performed. · Future work includes validation using representative clinical datasets.

Thank You

Questions?

Abubakar Amidu · 3MTT DeepTech Cohort 2 – DS/ML Mentorship · Abuja, Nigeria

Live Resources

GitHub Repository

<https://github.com/Ahamid/P001-MVR>



Live Streamlit Application

<https://p001-mvh-ia6ziomh6vr9thdamqgac.streamlit.app/>



Scan the QR codes to explore the complete project.